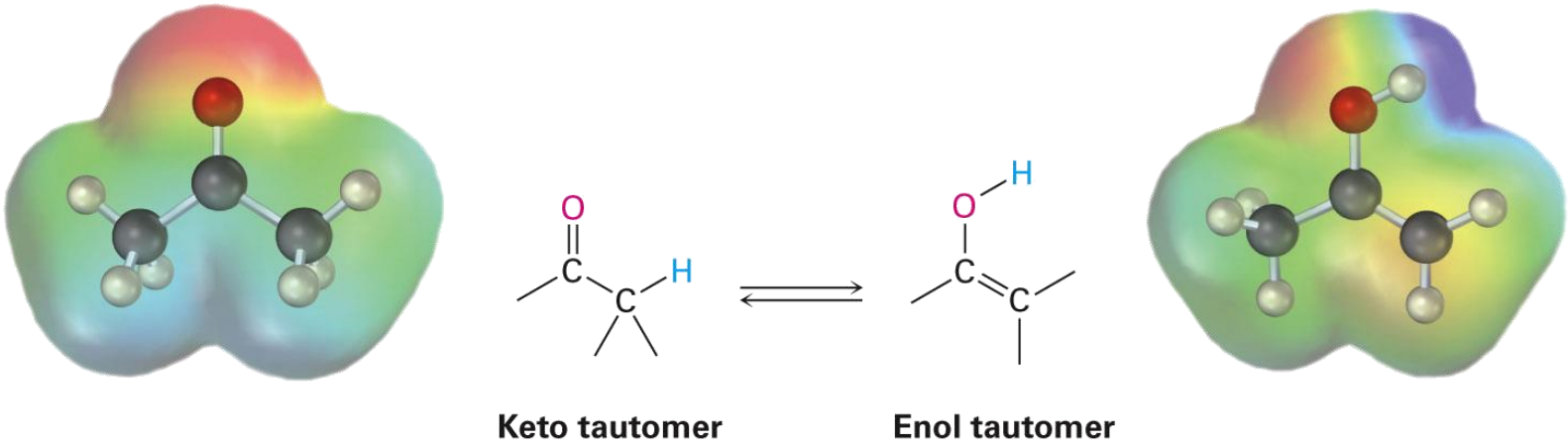
Chapter 22
Carbonyl Alpha-Substitution Reactions

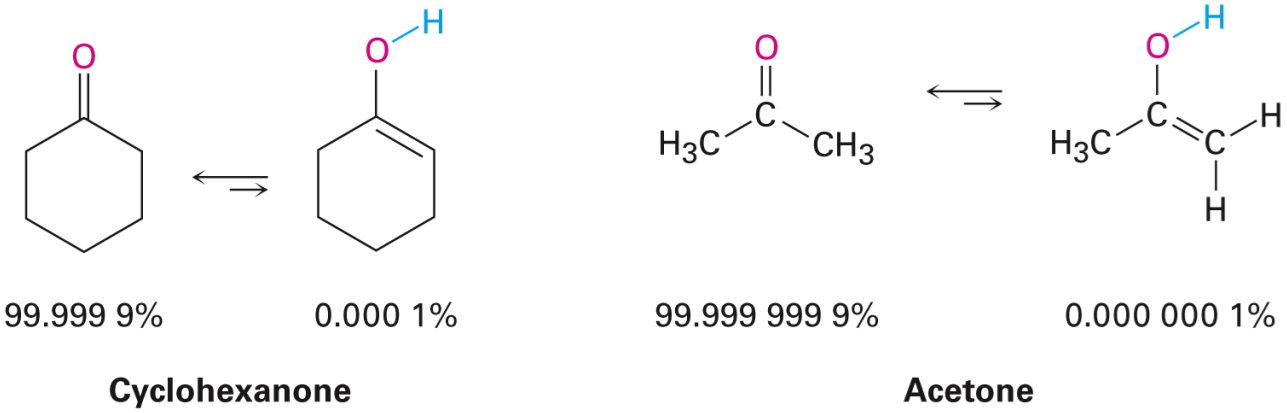
Keto-Enol Tautomerism

- A carbonyl compound with a hydrogen atom on its α carbon rapidly equilibrates with its corresponding **enol** isomer
- **Tautomers**: Isomers that interconvert spontaneously, usually with the change in position of a hydrogen



Keto-Enol Tautomerism

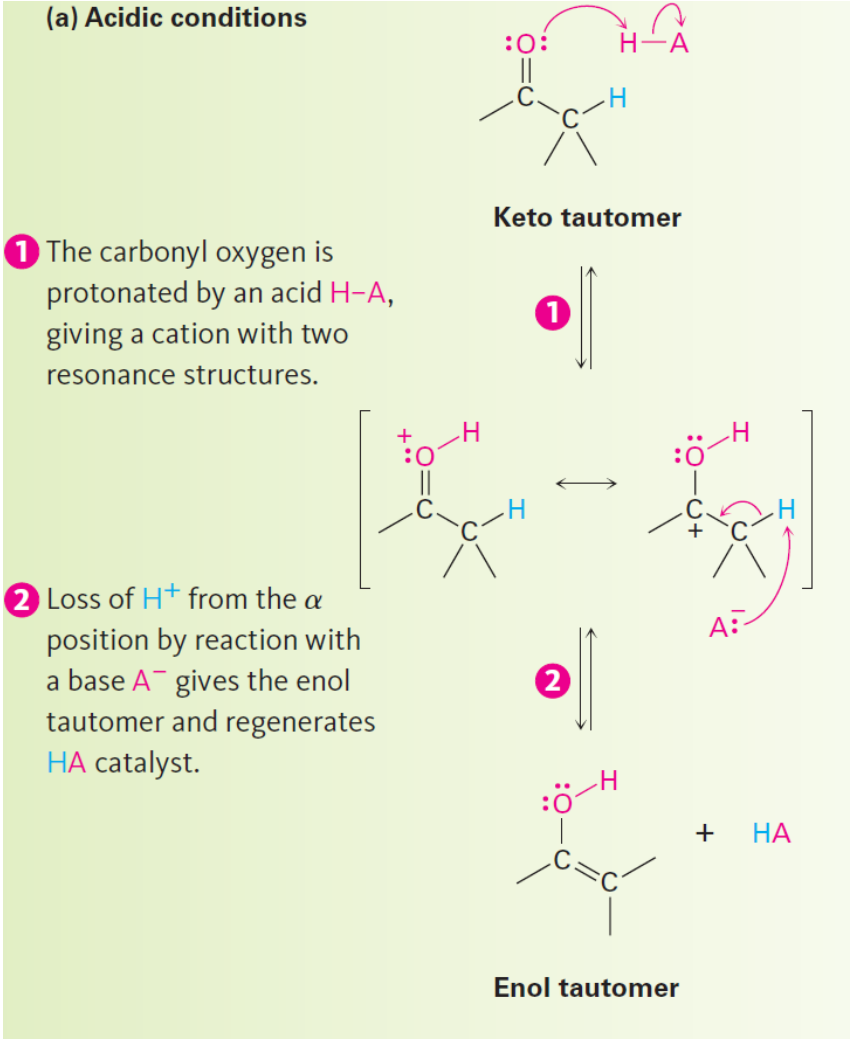
- Enol tautomer is often present in small extent and cannot be isolated easily
- Enols are responsible for much of the chemistry of carbonyl compounds



- Keto-enol tautomerism of carbonyl compounds is catalyzed by both acids and bases

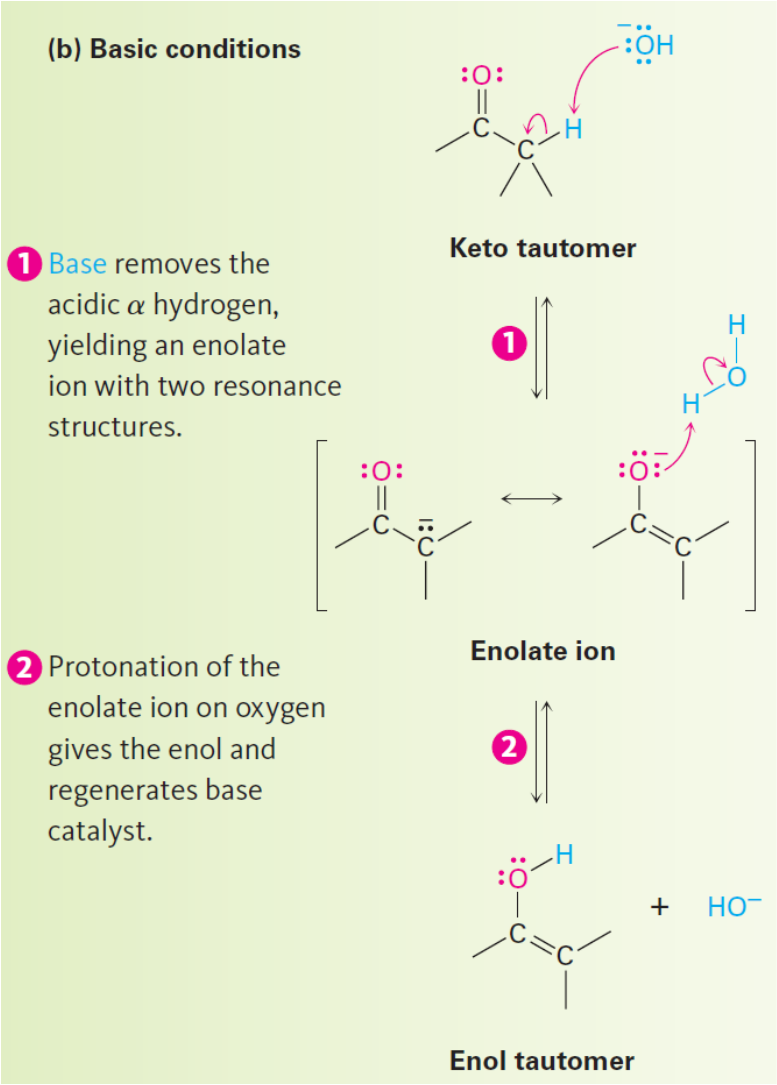
Keto-Enol Tautomerism

- Acid catalysis occurs due to protonation of carbonyl oxygen atom



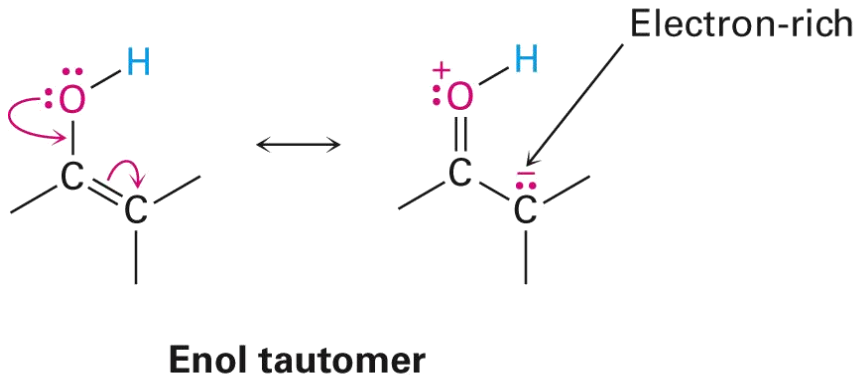
Keto-Enol Tautomerism

- Carbonyl compound can act as an acid and donate one of its *a* hydrogens to a sufficiently strong base, yielding an **enolate ion**

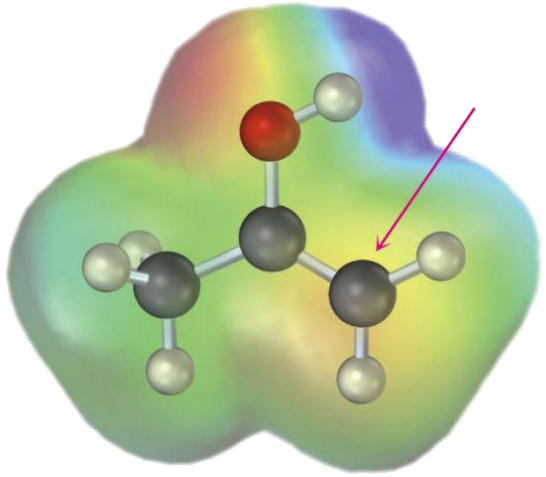


Reactivity of Enols: Alpha-Substitution Reactions

- Enols behave as nucleophiles and react with electrophiles
- Enols are more electron-rich and correspondingly more reactive than alkenes

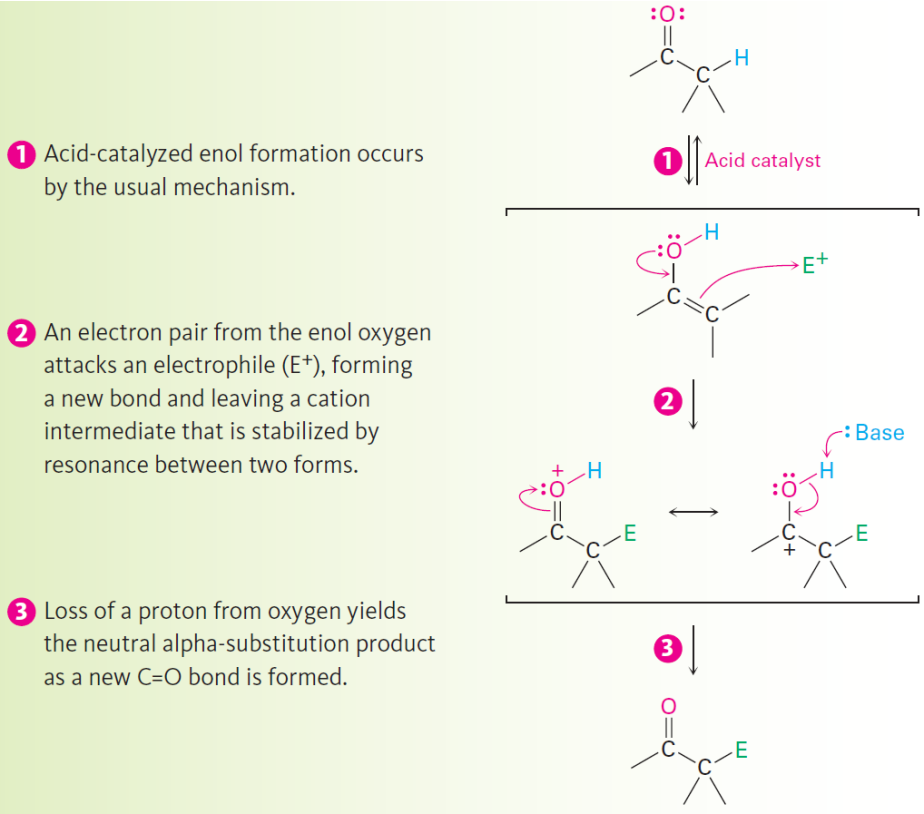


Enol tautomer



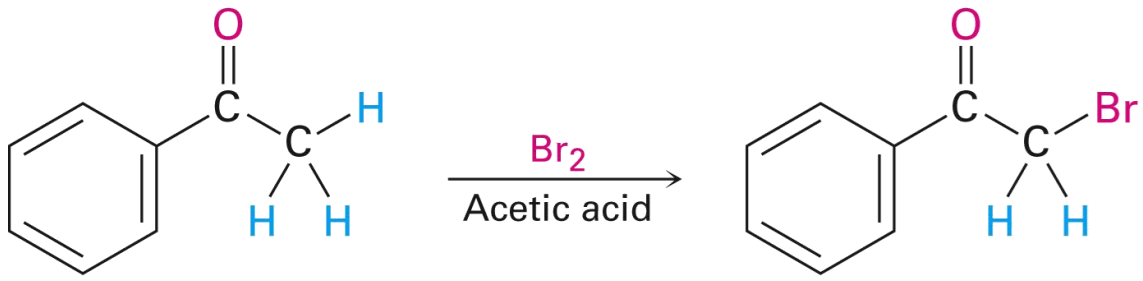
General Mechanism of Addition to Enols

- When an enol reacts with an electrophile the intermediate cation immediately loses the –OH proton to give an α-substituted carbonyl compound



Alpha Halogenation of Aldehydes and Ketones

- Aldehydes and ketones can be halogenated at their α positions by reaction with Cl_2, Br_2, or I_2 in acidic solution

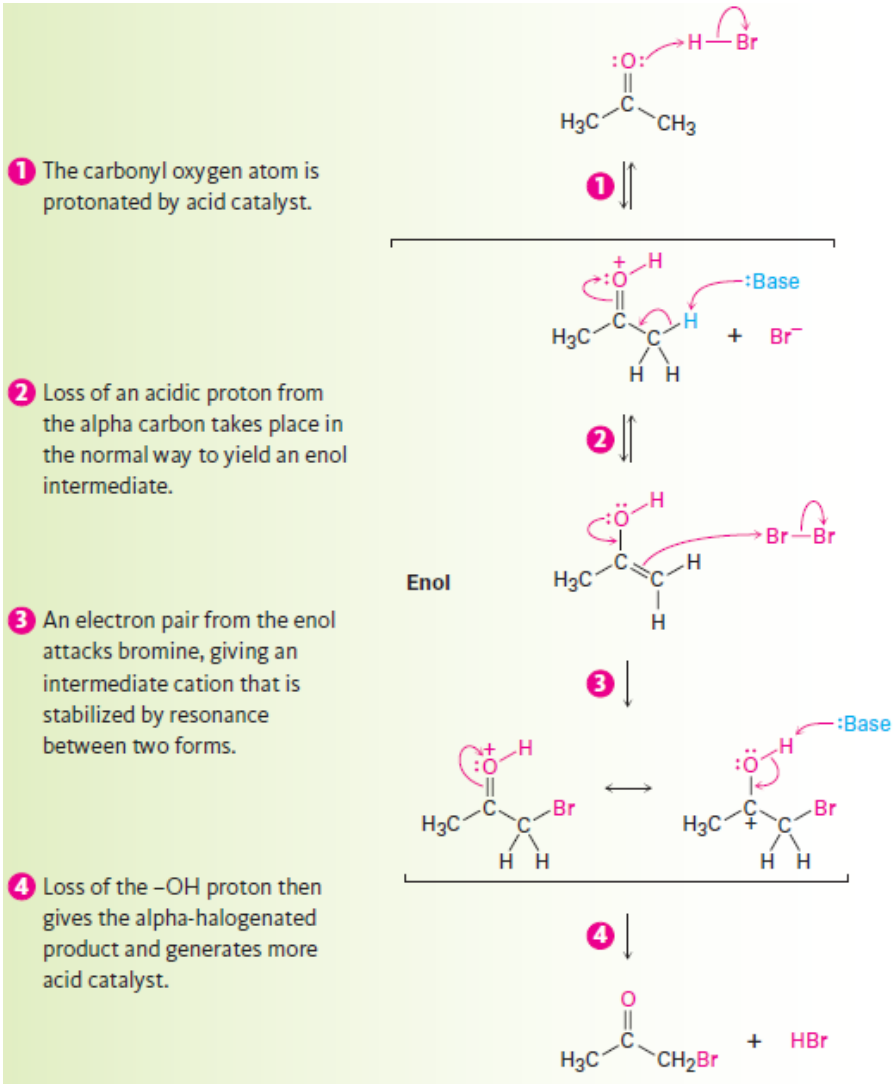


Acetophenone **α-Bromoacetophenone (72%)**

- Ketone halogenation also occur in biological systems

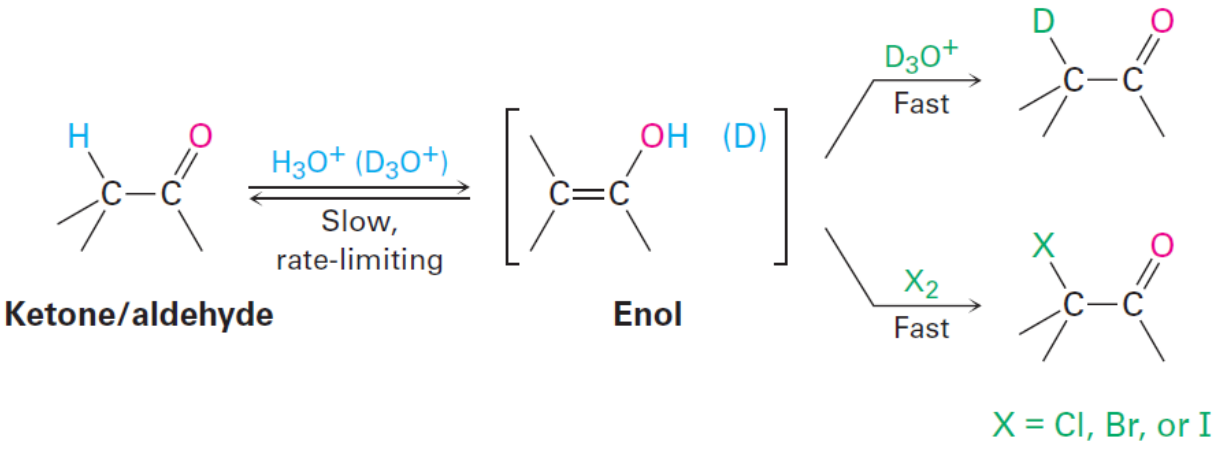
Alpha Halogenation of Aldehydes and Ketones

- α-substitution reaction is proceeded by acid-catalyzed formation of an enol intermediate



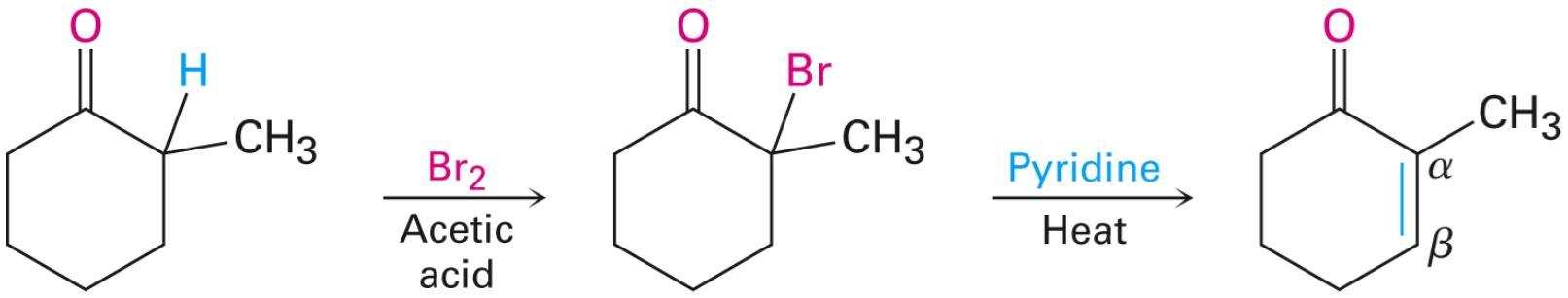
Alpha Halogenation of Aldehydes and Ketones

- The rate of halogenation is independent of the halogen's identity and concentration
- If an aldehyde or ketone is treated with D_3O^+, the α hydrogens are replaced by deuterium at the same rate as halogenation
  - Common intermediate is involved in both processes



Elimination Reactions of α-Bromoketones

- α-Bromo ketones can be dehydrobrominated by base treatment to yield α,β-unsaturated ketones



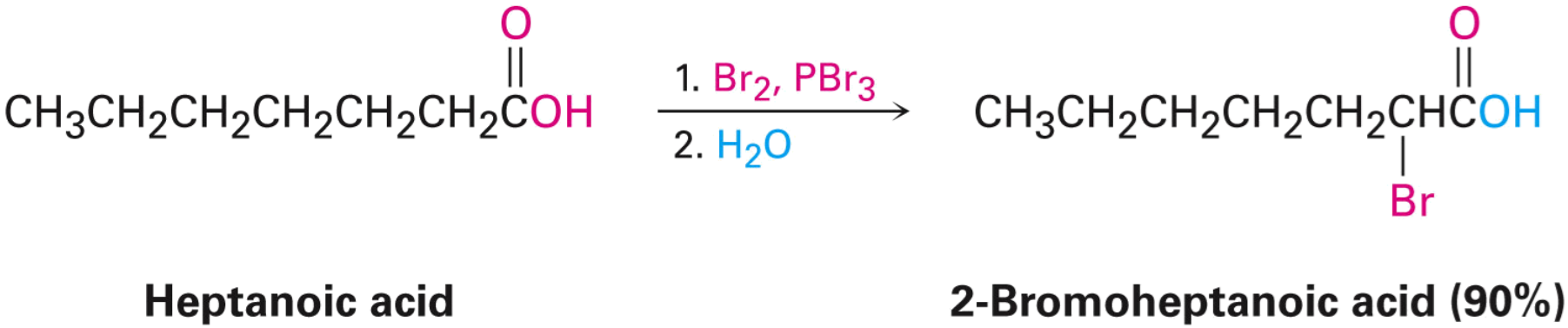
2-Methylcyclo-hexanone

2-Bromo-2-methyl-cyclohexanone

2-Methyl-2-cyclo-hexenone (63%)

Alpha Bromination of Carboxylic Acids

- Acids, esters, and amides do not react with Br_2
  - They are brominated by a mixture of Br_2 and PBr_3 (Hell–Volhard–Zelinskii reaction)

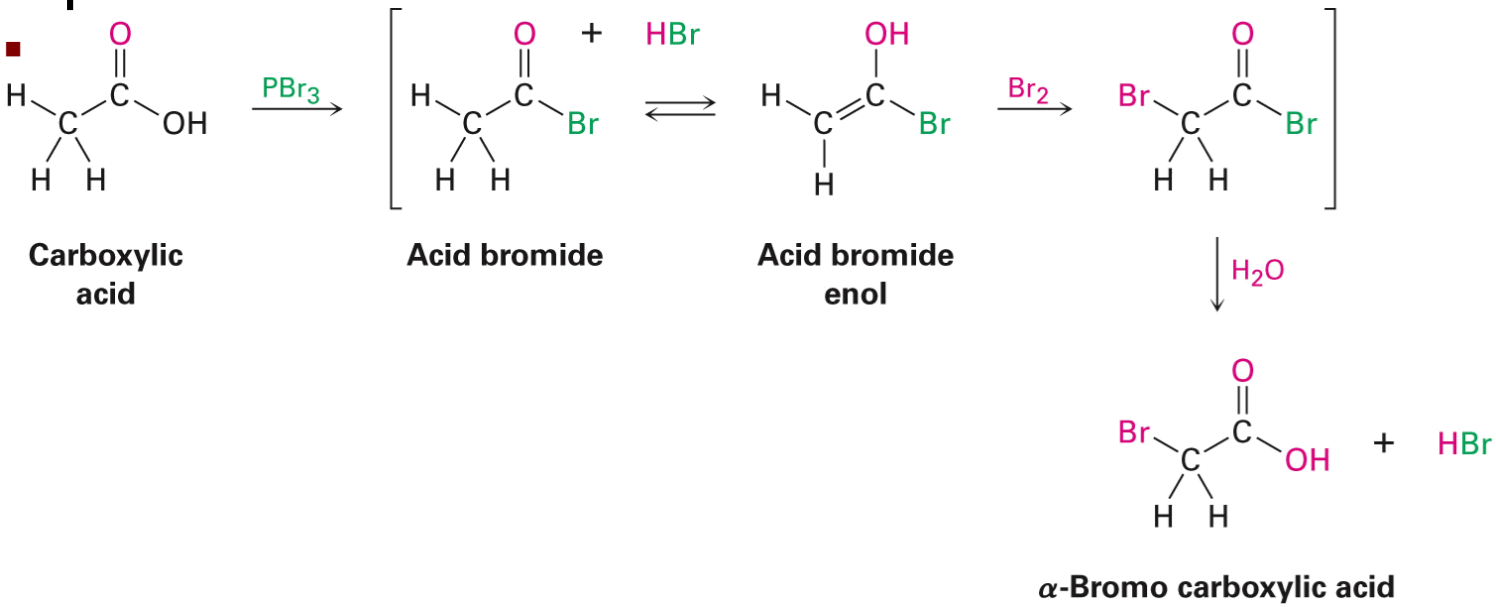


Heptanoic acid

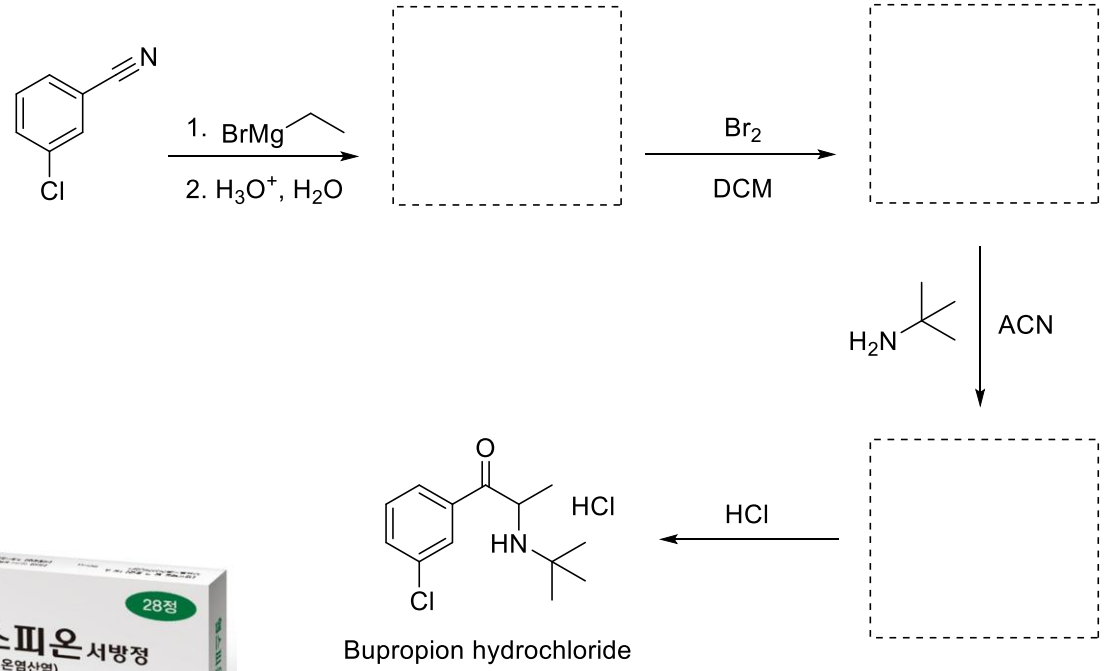
2-Bromoheptanoic acid (90%)

Alpha Bromination of Carboxylic Acids

- PBr_3 converts –COOH to –COBr
  - The resultant enol reacts with Br_2 to give α-bromo acid bromide
  - Water is used to hydrolyze the acid bromide in a nucleophilic acyl substitution reaction to yield product



Synthesis of bupropion hydrochloride



Bupropion hydrochloride

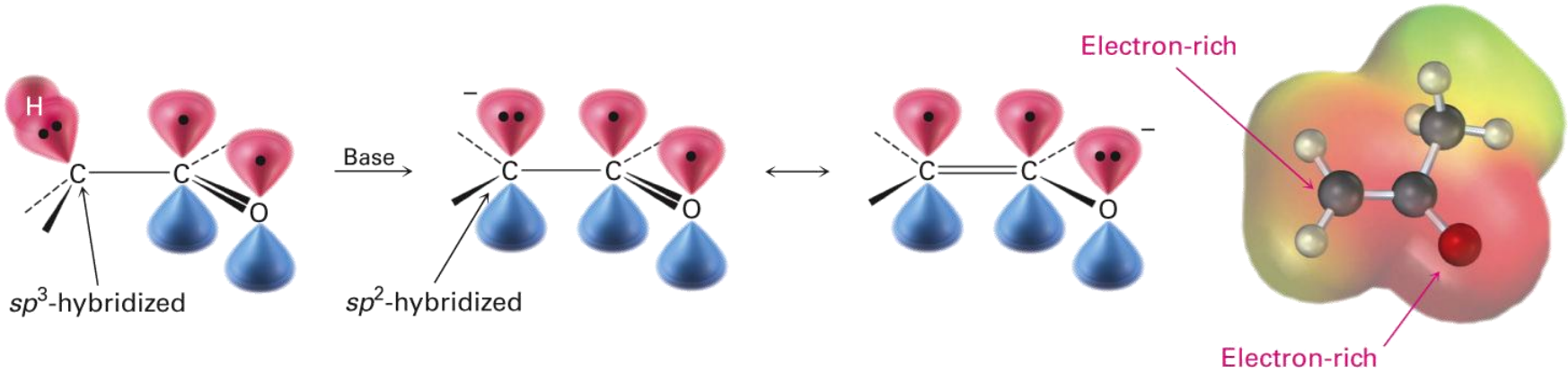


- 약물 계열: Aminoketone antidepressant
- 미국 최초 승인: 1985년
- 주요 적응증: 우울증 및 금연 보조 치료

Synthetic Communications, 40: 1566–1573, **2010**

Acidity of Alpha Hydrogen Atoms: Enolate Ion Formation

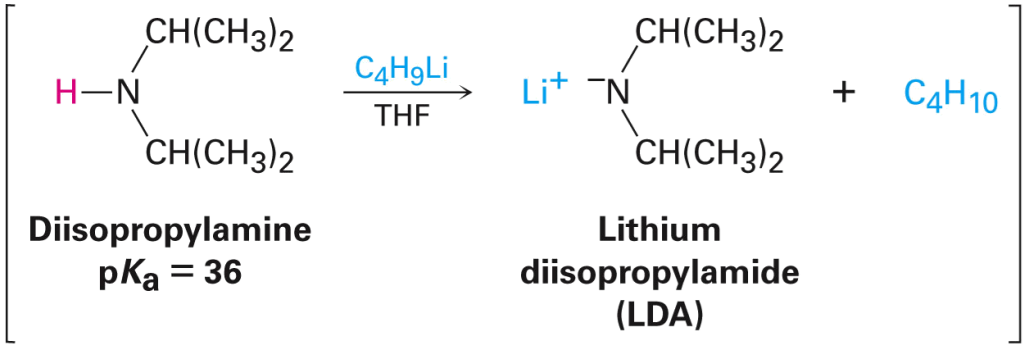
- Carbonyl compounds can act as weak acids
- Strong base is needed for enolate ion formation

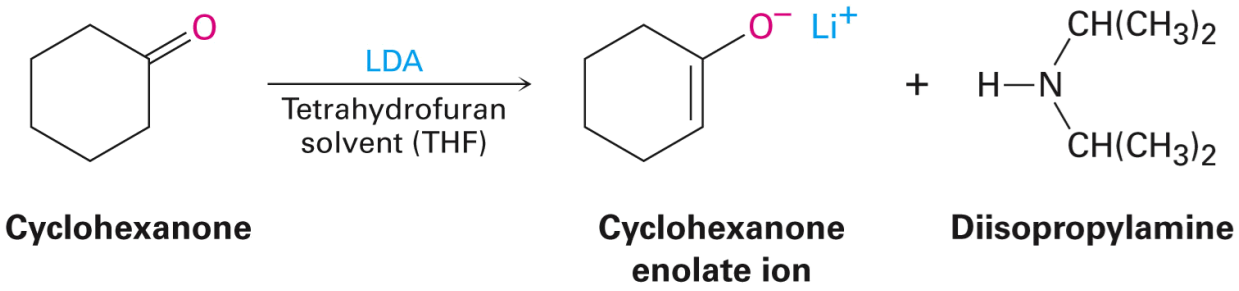


- Sodium hydride (NaH) or lithium diisopropylamide [LiN(i-$C_3H_7)_2$] (LDA) are strong enough to form the enolate

Acidity of Alpha Hydrogen Atoms: Enolate Ion Formation

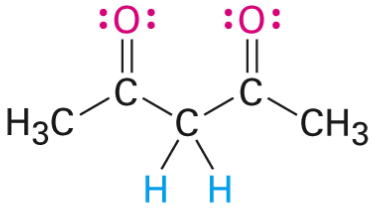
- LDA is from butyllithium (BuLi) and diisopropylamine (pK_a = 36)
- Soluble in organic solvents and effective at low temperature with many compounds





Acidity of Alpha Hydrogen Atoms: Enolate Ion Formation

- When a hydrogen atom is flanked by two carbonyl groups, its acidity is enhanced
- Negative charge of enolate delocalizes over both carbonyl groups



2,4-Pentanedione (pK_a = 9)

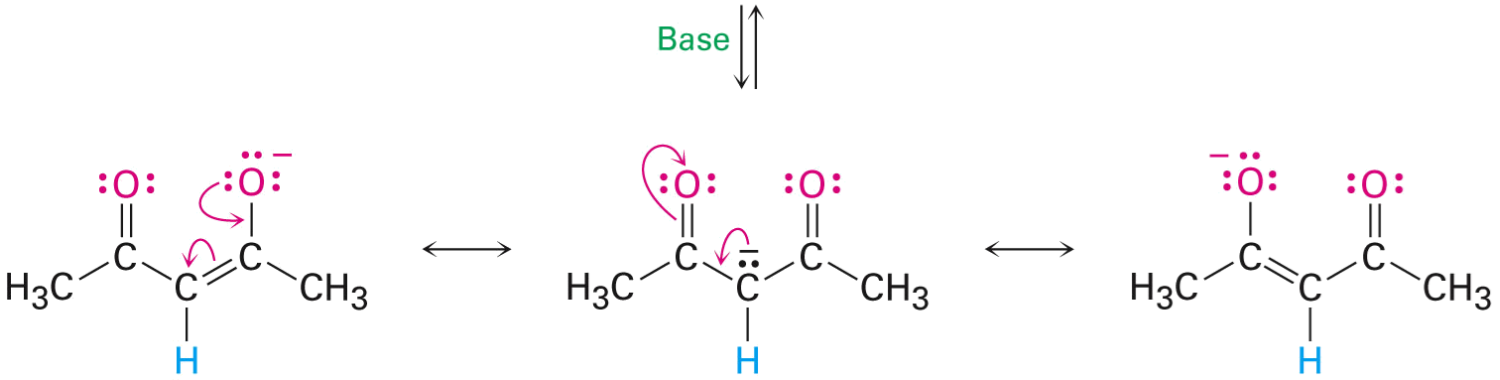
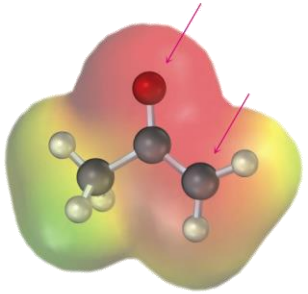


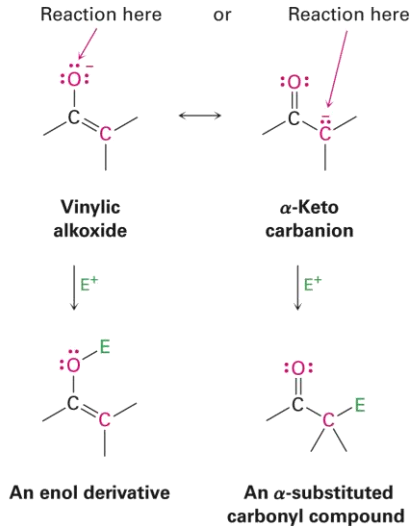
Table 22.1 - Acidity Constants for Some Organic Compounds

Functional group	Example	pK_a	Functional group	Example	pK_a
Carboxylic acid	CH_3COH	5	Aldehyde	CH_3CH	17
1,3-Diketone	$CH_3CCH_2CCH_3$	9	Ketone	CH_3CCH_3	19
3-Keto ester	$CH_3CCH_2COCH_3$	11	Thioester	CH_3CSCH_3	21
1,3-Diester	$CH_3OCCH_2COCH_3$	13	Ester	CH_3COCH_3	25
Alcohol	CH_3OH	16	Nitrile	$CH_3C{\equiv}N$	25
Acid chloride	CH_3CCl	16	*N*,*N*-Dialkylamide	$CH_3CN(CH_3)_2$	30
			Dialkylamine	$HN(i\text{-}C_3H_7)_2$	36

Reactivity of Enolate Ions

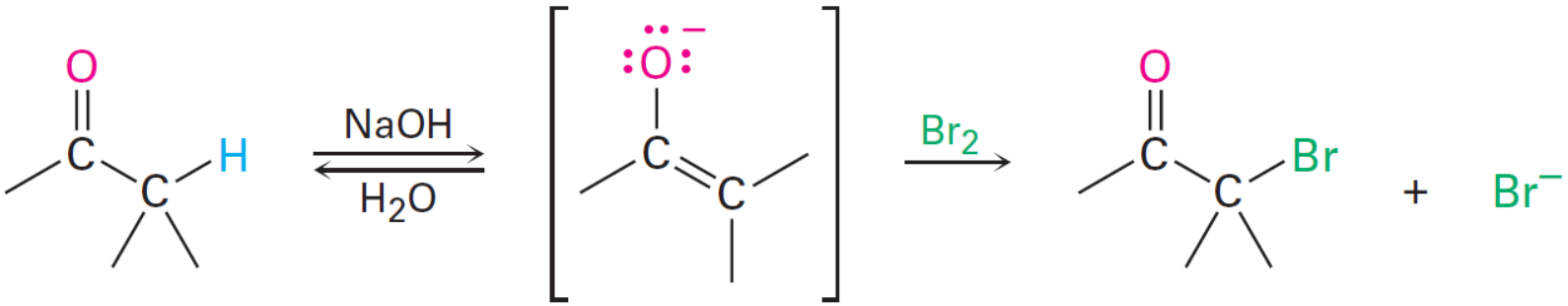
- Enolate ions can be looked at either as vinylic alkoxides (C=C–O^-) or as *α*-keto carbanions ($^-$C–C=O)
  - Enolate ions can react with electrophiles
- Reaction on oxygen yields an enol derivative
- Reaction on carbon yields an *α*-substituted carbonyl compound





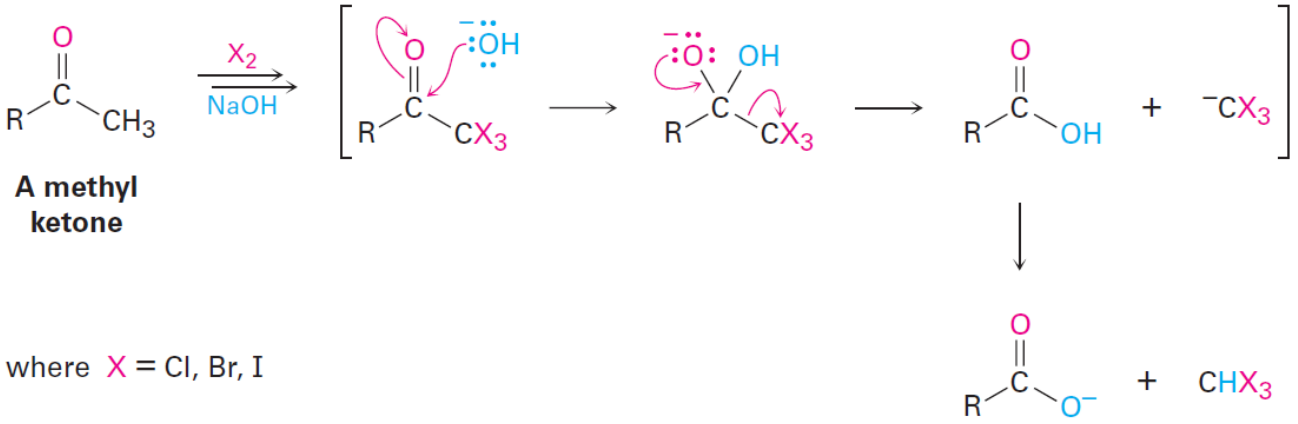
Reactivity of Enolate Ions

- Aldehydes and ketones undergo base-promoted α halogenation
- Weak bases are effective for halogenation because it is not necessary to convert the ketone completely into its enolate ion



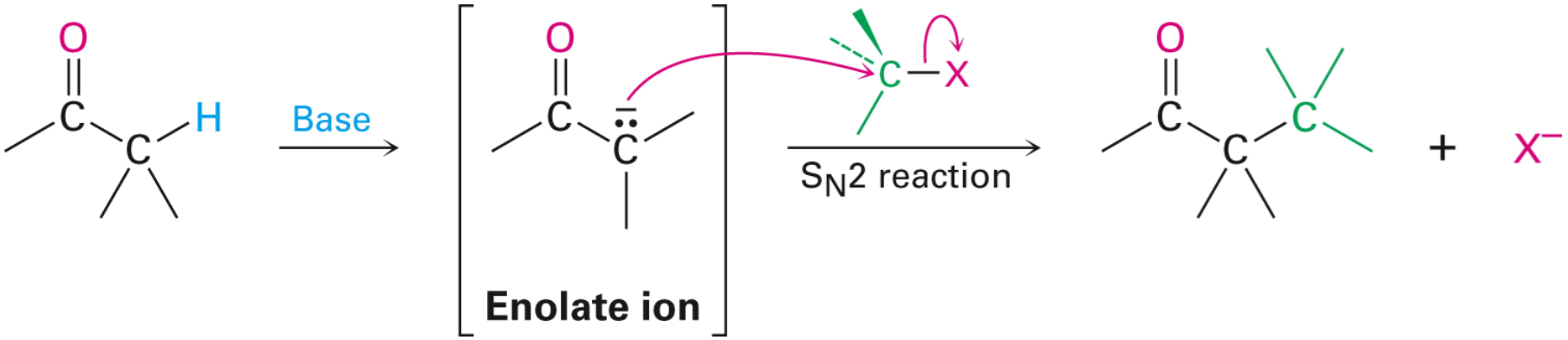
Reactivity of Enolate Ions

- Base-promoted halogenation of aldehydes and ketones is seldom used
- If excess base and halogen are used, a methyl ketone is triply halogenated and then cleaved by base in the haloform reaction
  - A halogen-stabilized carbanion acts as a leaving group



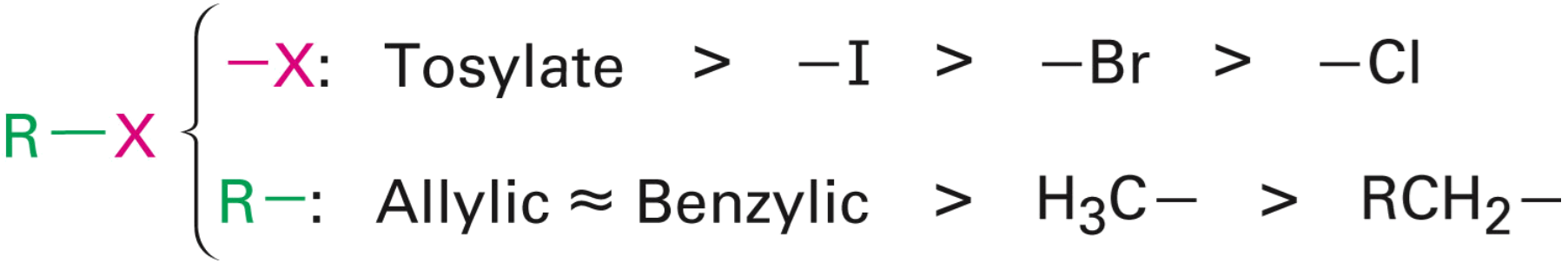
Alkylation of Enolate Ions

- Base-promoted reaction occurs through an enolate ion intermediate



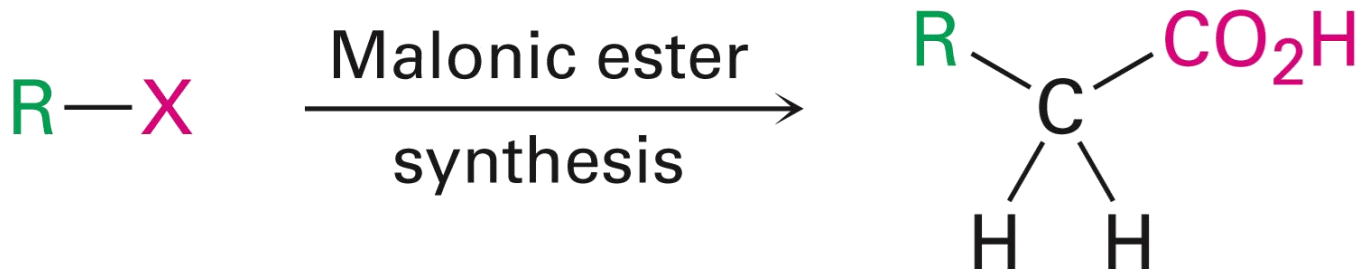
Constraints on Enolate Alkylation

- S_N2 reaction - Leaving group X can be chloride, bromide, iodide, or tosylate
- R should be primary or methyl and preferably should be allylic or benzylic
- Secondary halides react poorly, and tertiary halides don't react at all because of competing elimination



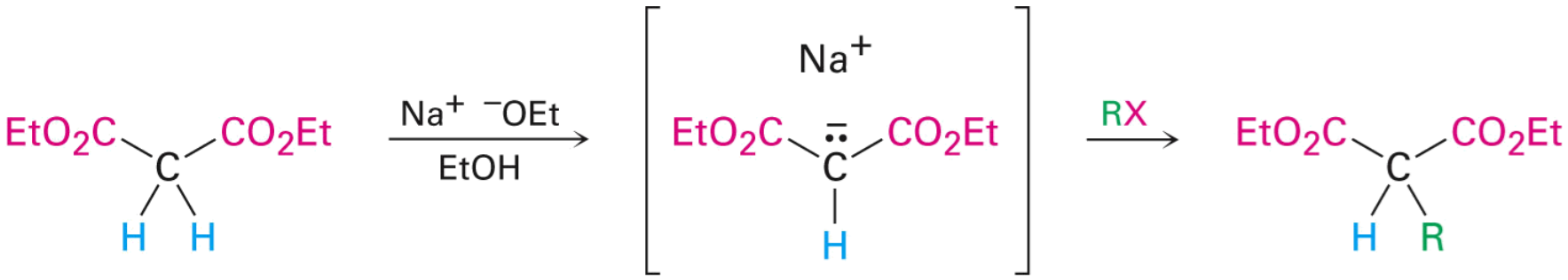
Malonic Ester Synthesis

- For preparing a carboxylic acid from an alkyl halide while lengthening the carbon chain by two atoms



Formation of Enolate and Alkylation

- Malonic ester (diethyl propanedioate) is easily converted into its enolate ion by reaction with sodium ethoxide in ethanol
- The enolate is a good nucleophile that reacts rapidly with an alkyl halide to give an *α*-substituted malonic ester



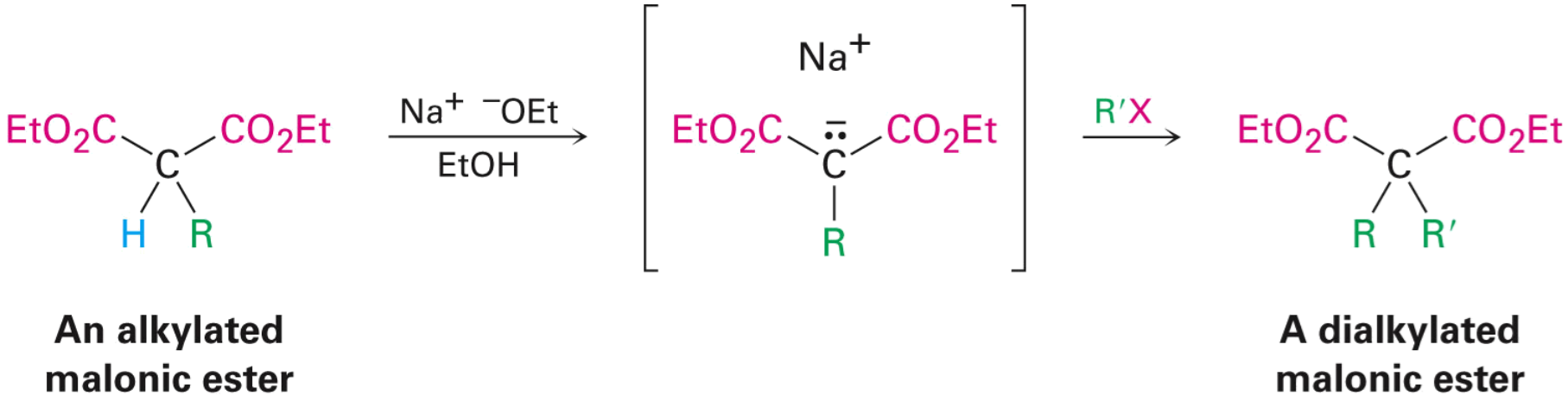
Diethyl propanedioate (malonic ester)

Sodio malonic ester

An alkylated malonic ester

Dialkylation

- The product has an acidic α-hydrogen, allowing the alkylation process to be repeated

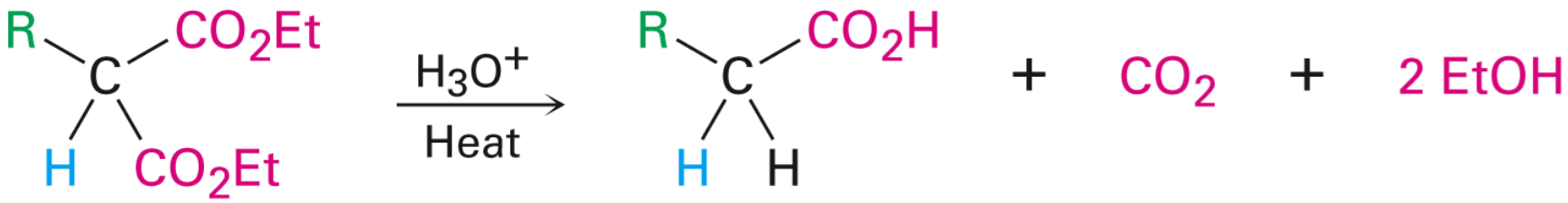


An alkylated malonic ester

A dialkylated malonic ester

Hydrolysis and Decarboxylation

- The malonic ester derivative hydrolyzes in acid and loses CO_2 (decarboxylation) to yield a substituted monoacid

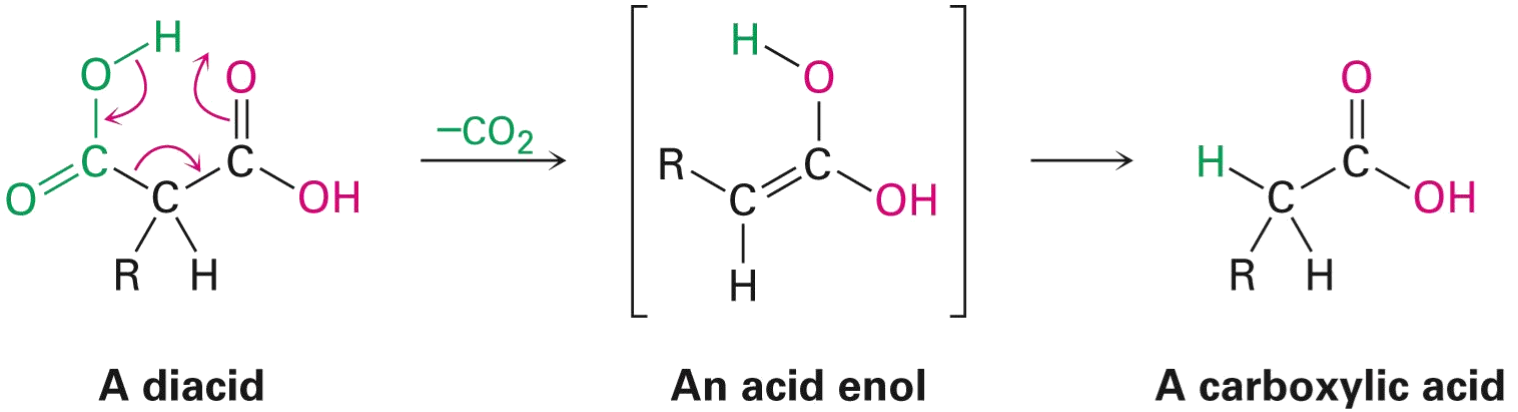


An alkylated malonic ester

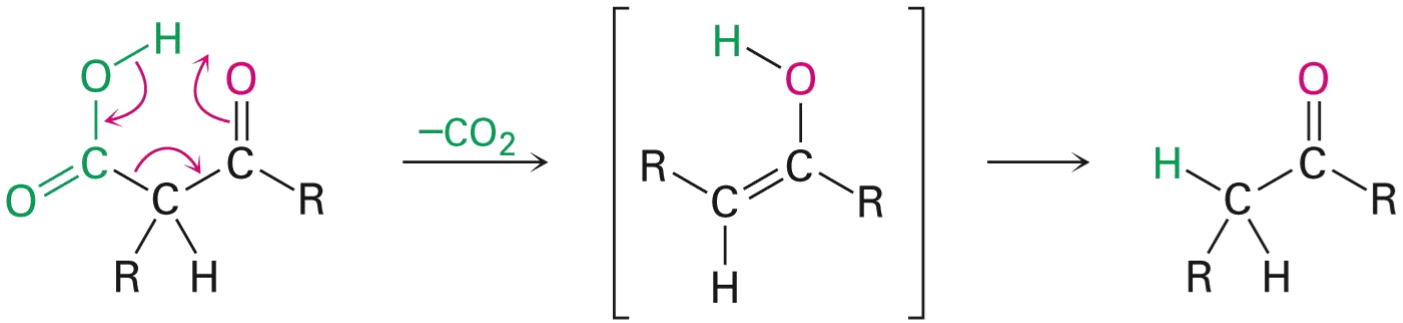
A carboxylic acid

Decarboxylation of β-Ketoacids

- Decarboxylation requires a carbonyl group two atoms away from the $-CO_2H$



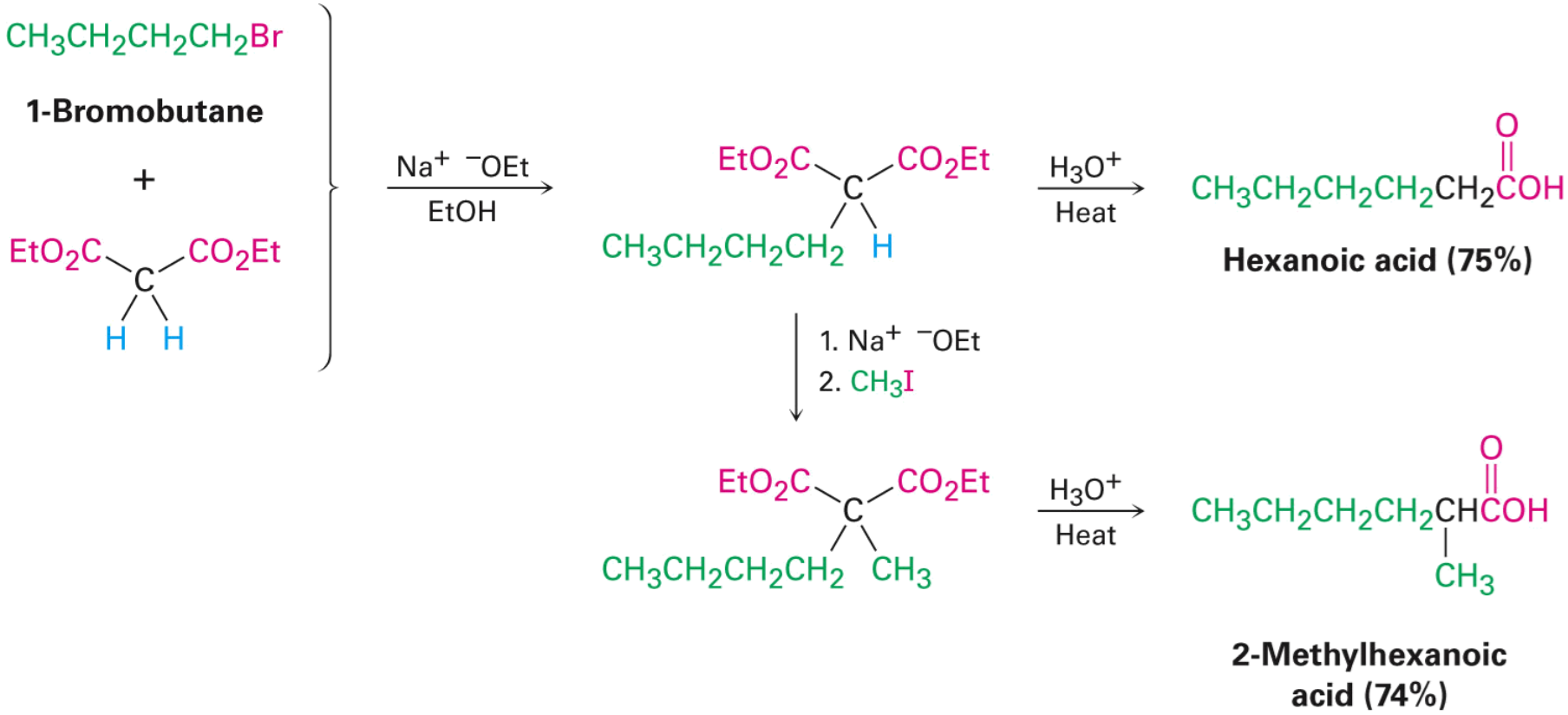
A diacid **An acid enol** **A carboxylic acid**



A *β*-keto acid **An enol** **A ketone**

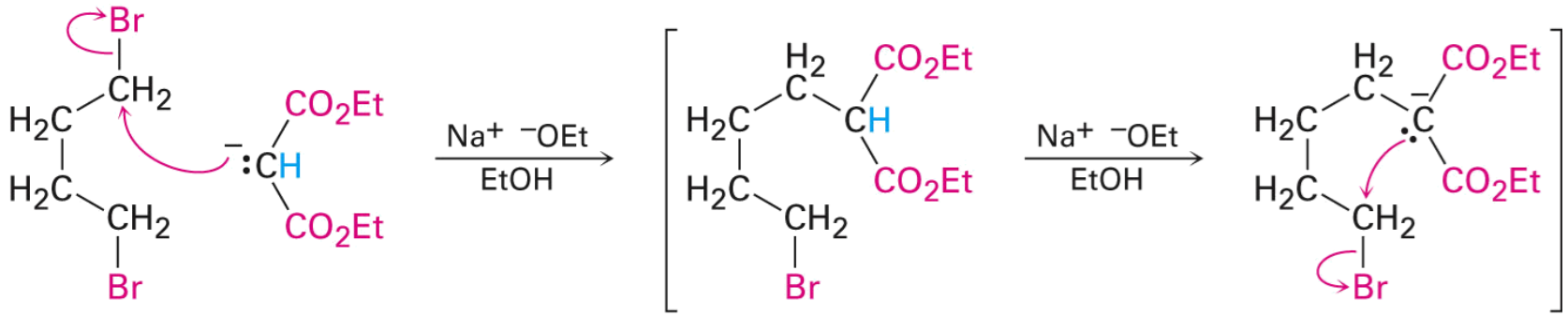
Overall Conversion

- The malonic ester synthesis converts an alkyl halide into a carboxylic acid while lengthening the carbon chain by two atoms

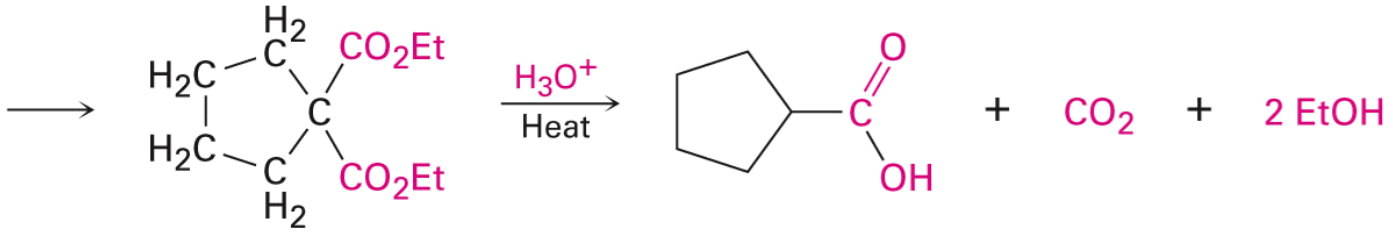


Preparation of Cycloalkane Carboxylic Acids

- 1,4-dibromobutane reacts twice, giving a cyclic product
- Three-, four-, five-, and six-membered rings can be prepared in this way



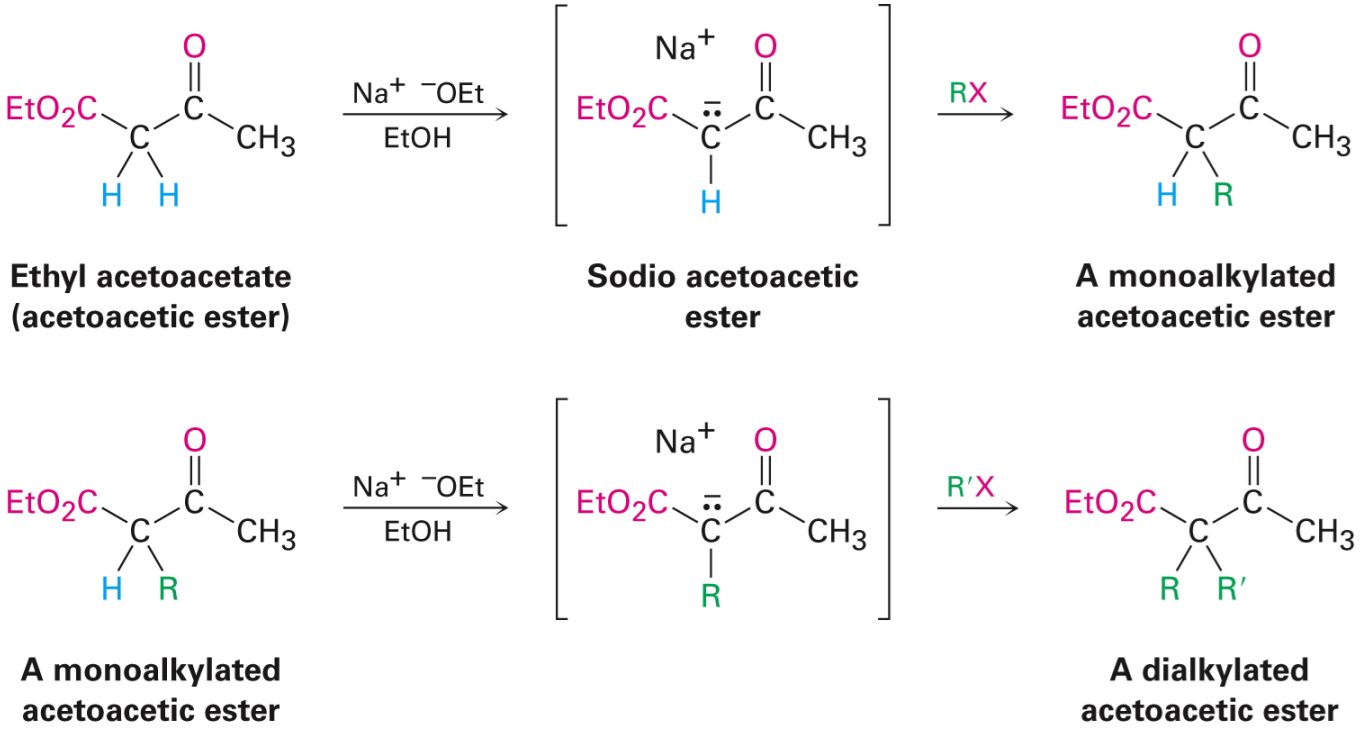
1,4-Dibromobutane



Cyclopentane-carboxylic acid

Acetoacetic Ester (Ethyl Acetoacetate)

- α carbon is flanked by two carbonyl groups, so it readily becomes an enolate ion
- This can be alkylated by an alkyl halide and also can react with a second alkyl halide



Ethyl acetoacetate (acetoacetic ester)

Sodio acetoacetic ester

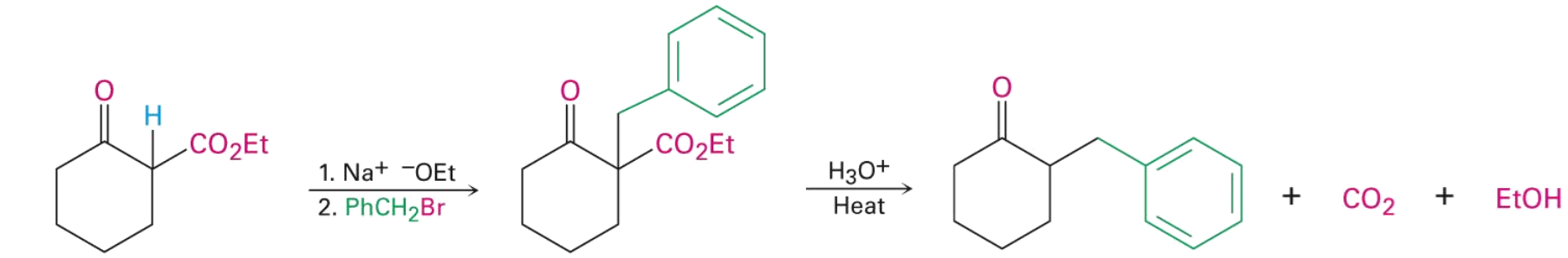
A monoalkylated acetoacetic ester

A monoalkylated acetoacetic ester

A dialkylated acetoacetic ester

Generalization: β-Keto Esters

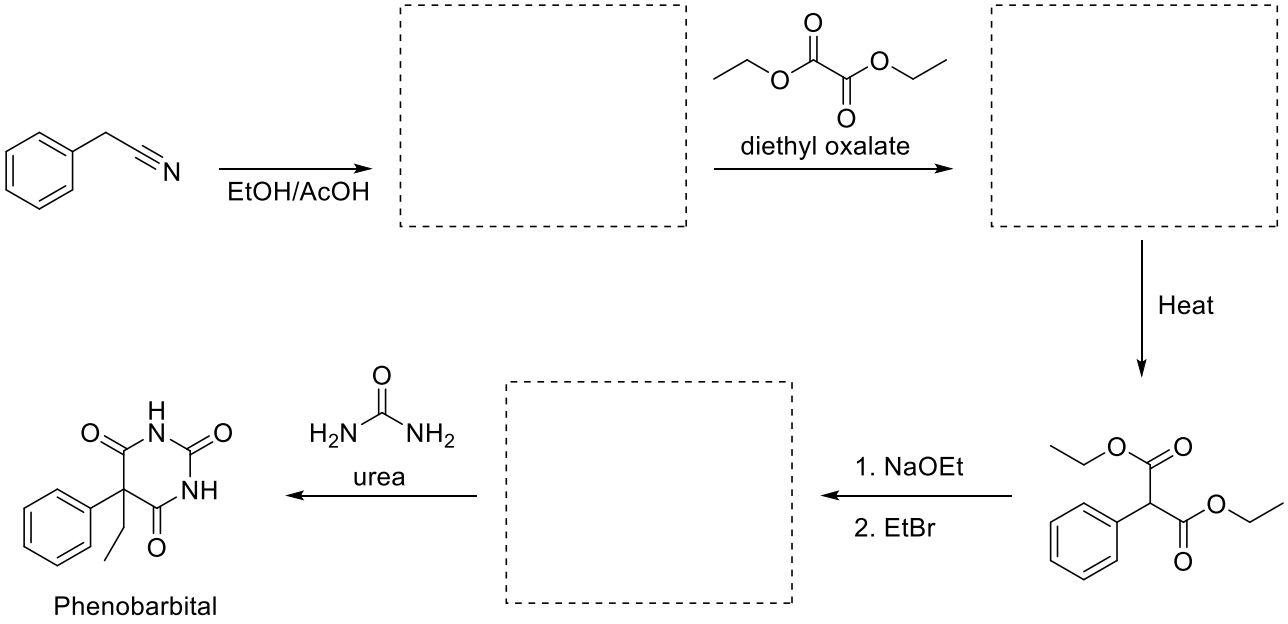
- Sequence
  - Enolate ion formation
  - Alkylation
  - Hydrolysis/decarboxylation
- Cyclic β-keto esters give 2-substituted cyclohexanones



Ethyl 2-oxocyclohexane-carboxylate (a cyclic β-keto ester)

2-Benzylcyclohexanone (77%)

Synthesis of phenobarbital





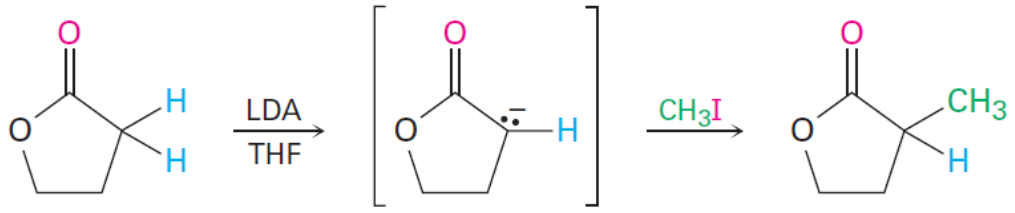
- 최초 시판: 1912년
- 약물 계열: Barbiturate
- 주요 용도: 항경련제
- 식품의약품안전처(MFDS)
- FDA 승인 제형: Sezaby (phenobarbital sodium 주사제)

https://commons.wikimedia.org/wiki/File:Phenobarbital_synthesis.png

Direct Alkylation of Ketones, Esters, and Nitriles

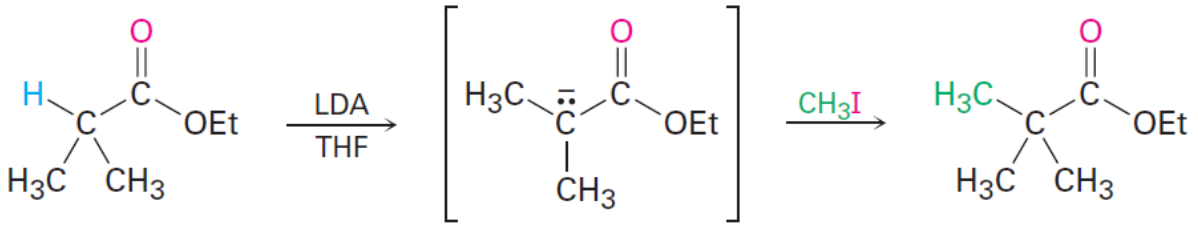
- Ketones, esters, and nitriles can all be alkylated using LDA or related dialkylamide bases in THF

Lactone



Butyrolactone **2-Methylbutyrolactone (88%)**

Ester



Ethyl 2-methylpropanoate **Ethyl 2,2-dimethylpropanoate (87%)**

Direct Alkylation of Ketones, Esters, and Nitriles

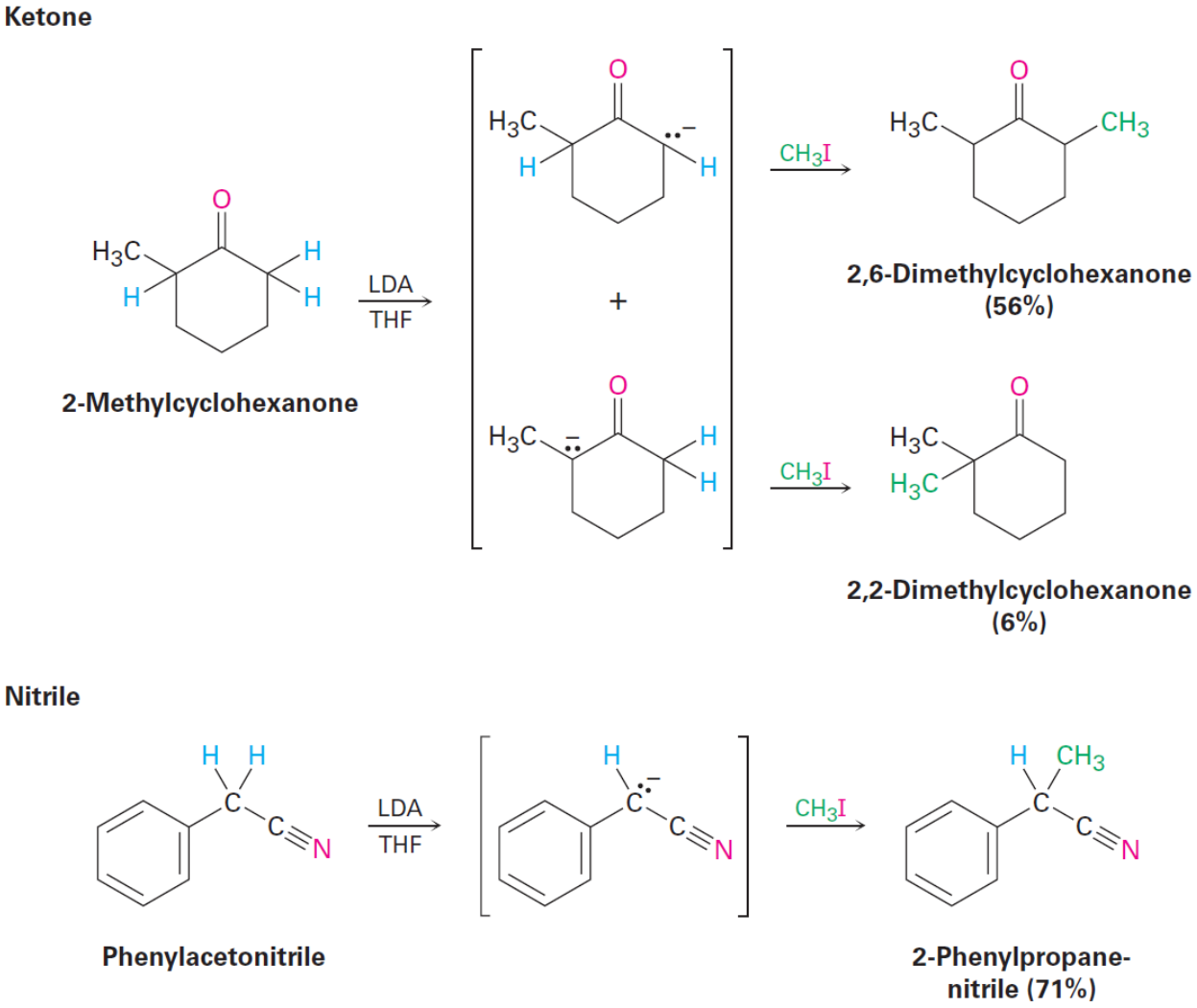
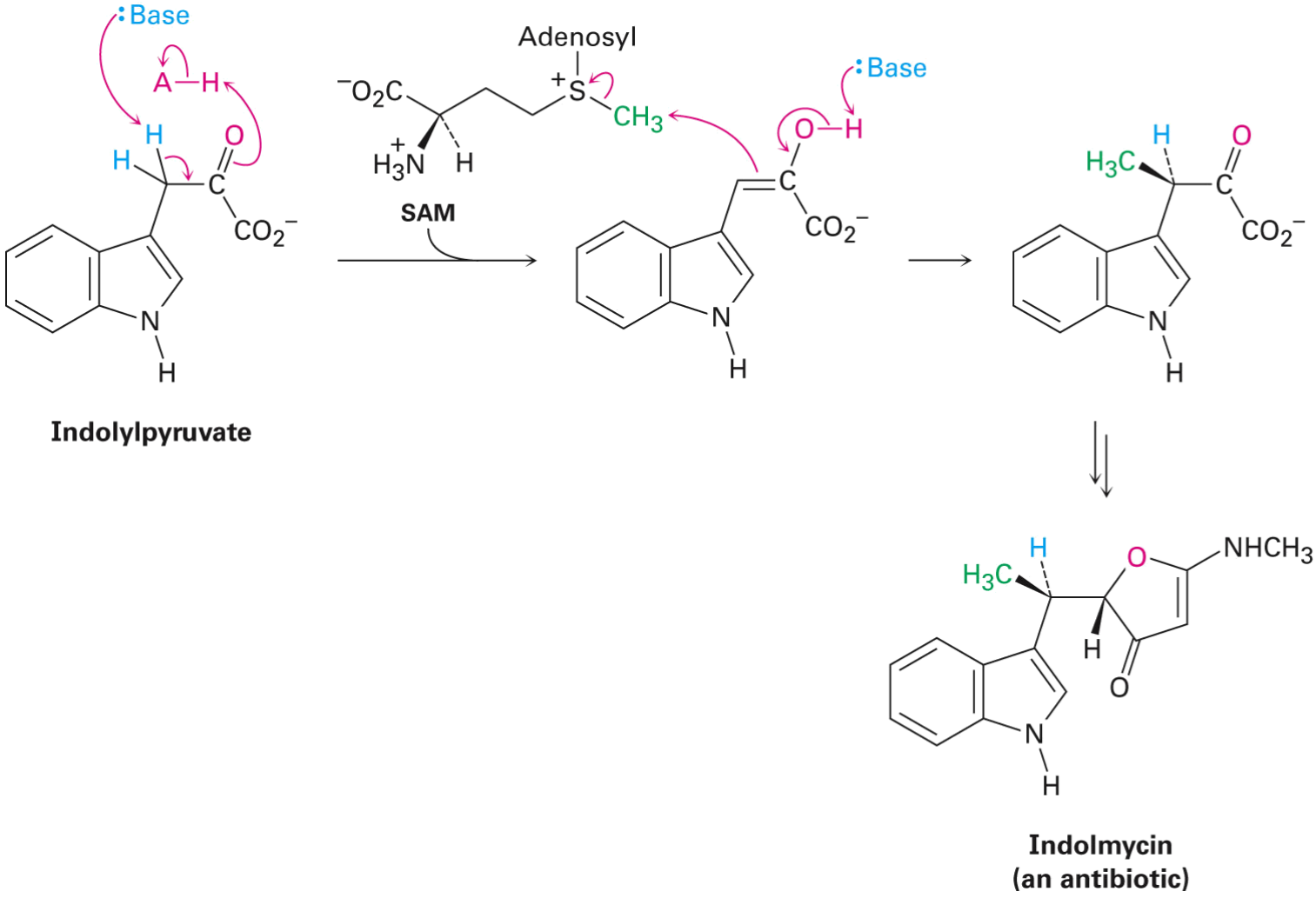
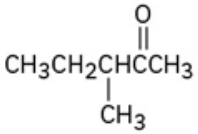
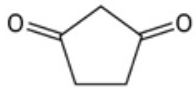


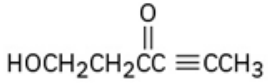
Figure 22.6 - Biosynthesis of Indolmycin from Indolylpyruvate

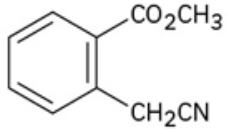


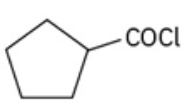
PROBLEM 22-37 Identify all the acidic hydrogens ($pK_a < 25$) in the following molecules:

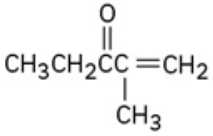
(a) 

(b) 

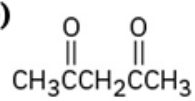
(c) 

(d) 

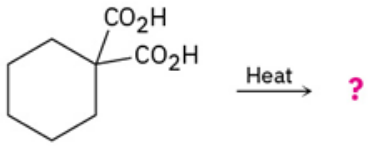
(e) 

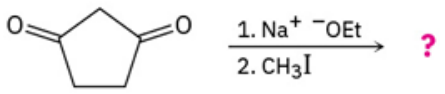
(f) 

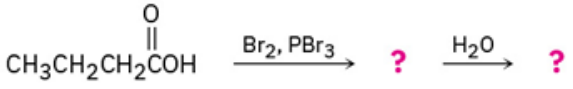
PROBLEM 22-38 Rank the following compounds in order of increasing acidity:

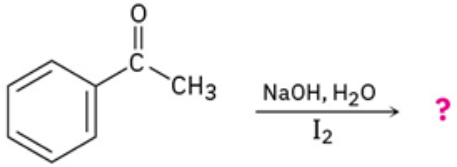
(a) $CH_3CH_2CO_2H$ **(b)** CH_3CH_2OH **(c)** $(CH_3CH_2)_2NH$ **(d)** CH_3COCH_3 **(e)**  **(f)** CCl_3CO_2H

PROBLEM 22-42 Predict the product(s) of the following reactions:

(a) 

(b) 

(c) 

(d) 

PROBLEM 22-54 Fill in the reagents **a**–**c** that are missing from the following scheme:

